

# Dynamic Mode Decomposition (DMD)

Aim: Learn linear models from snapshot data.

Recap Given snapshot matrices

$$X = \begin{bmatrix} 1 & & 1 \\ x_1 & \dots & x_m \\ 1 & & 1 \end{bmatrix} \quad \text{and} \quad Y = \begin{bmatrix} 1 & & 1 \\ y_1 & \dots & y_m \\ 1 & & 1 \end{bmatrix},$$

$d \times m$                        $d \times m$

where  $y_j = \bar{T} x_j \quad j=1, \dots, m,$

DMD constructs the best linear fit

$$(*) \quad \hat{T} = \underset{S \in \mathbb{R}^{d \times d}}{\operatorname{argmin}} \|Y - SX\|.$$

$\Rightarrow$  Eigenvalues / Eigenvectors of  $\hat{T}$  give insight into "discrete-time" dynamics.

$\Rightarrow$  When  $m > d$  and  $\|\cdot\| = \|\cdot\|_2$ , (\*) is an overdetermined least-squares problem for  $\hat{T}$  (see Lectures 5-6).



# DMD for High-Dimensional Dynamics

DMD is often applied to dynamical systems with high-dimensional state-space and relatively small number of snapshots. Here,  $m \ll d$  and the LS-problem is underdetermined. There is no unique soln!

$$\underset{s \in \mathbb{R}^d}{\text{argmin}} \left\| \begin{bmatrix} x_0 \\ \vdots \\ x_m \end{bmatrix} S - \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} \right\|$$

If  $\text{rank}(X) = m$ ,  $\infty$ -ly many solutions.

Q: Which soln should we pick?

Example: Flow past a cylinder





Snapshots correspond to velocity (and sometimes pressure) field measurements on a fine grid of pixels discretizing the flow field

$$\begin{bmatrix} | & & | \\ y_1 & \dots & y_m \\ | & & | \end{bmatrix} \approx \begin{bmatrix} & & \\ T & & \\ & & \\ dxd & & \end{bmatrix} \begin{bmatrix} | & & | \\ x_1 & \dots & x_m \\ | & & | \end{bmatrix}$$

↑ each column corresponds to velocity data at each pixel of the spatial discretization

⇒ The problem for  $T$  is underdetermined.

⇒ Typical for problems involving discrete measurements with complex spatial variations.

### Minimal Norm Solution

A natural place to start when asking "which" solution is a/the physical one is the minimal norm solution.



Given  $X, Y \in \mathbb{R}^{d \times m}$ ,  $\text{rank}(X) = m$ , consider

$$(44) \quad \hat{T} = \min_{S \in \mathbb{R}^{d \times d}} \|S\|_F \quad \text{s.t.} \quad Y = SX.$$

where  $\|S\|_F = \sqrt{\sum_{i,j} S_{ij}^2}$  is the Frobenius norm.

$\Rightarrow$  In some cases, minimal norm solns are motivated by variational properties of the underlying physical trajectories.

$\Rightarrow$  In other cases, the norm may act as a proxy for complexity, and the minimal norm soln is the "simplest."

$\Rightarrow$  The minimal norm solution may not always correspond to the "most physical" soln! See Lecture 13.

Q: How do we construct  $\hat{T}$  from the snapshot matrices  $X$  and  $Y$ ?



Claim: If  $\text{rank}(X) = m$ , then  $(\star\star)$  has a unique solution

$$\hat{T} = YX^+, \quad \text{where} \quad X^+ = (X^T X)^{-1} X^T$$

is the pseudoinverse of  $X$ .

PF First, since  $\text{rank}(X) = m$ ,  $X^T X$  is invertible and  $\hat{T}X = Y(X^T X)^{-1} X^T X = Y$ .  
Therefore,  $\hat{T}$  satisfies  $Y = \hat{T}X$ .

Second, we need to show that  $\hat{T}$  is the minimal norm solution of  $Y = SX$ . The general solution to  $Y = SX$  has form

$$(1) \quad S = Y(X^T X)^{-1} X^T + N$$

"particular" =  $\hat{T}$       "homogeneous"

where  $N \in \mathbb{R}^{\text{dod}}$  is any matrix s.t.  $0 = NX$ .  
I.e., the rows of  $N$  are orthogonal to the columns of  $X$ . We can write the Frob. norm  $\|S\|_F$  in "inner product" form:



$$\|S\|_F^2 = \text{trace } S^T S = \sum_{i=1}^d (S^T S)_{ii} = \sum_{i,j=1}^d S_{ij}^2.$$

Expanding  $\|S\|_F^2$  using (1), we get

$$\|S\|_F^2 = \text{trace} [(\hat{\tau} + N)^T (\hat{\tau} + N)]$$

$$= \text{trace} [\hat{\tau}^T \hat{\tau} + \hat{\tau}^T N + N^T \hat{\tau} + N^T N]$$

by linearity  
of trace

$$= \|\hat{\tau}\|_F^2 + \|N\|_F^2 + \text{trace} [\hat{\tau}^T N + N^T \hat{\tau}]$$

Now, since  $\text{trace}(AB) = \text{trace}(BA)$  (this is called the cyclical property of the trace)

$$\text{trace} [\hat{\tau}^T N + N^T \hat{\tau}] = \text{trace} [N \hat{\tau}^T] + \text{trace} [\hat{\tau} N^T]$$

$$\Rightarrow \text{trace} [N \hat{\tau}^T] = N X (X^T X)^{-1} Y^T = 0$$

$$\Rightarrow \text{trace} [\hat{\tau} N^T] = Y (X^T X)^{-1} X^T N^T = 0$$

$$= \underbrace{(N X)^T}$$

So  $\|S\|_F^2 = \|\hat{\tau}\|_F^2 + \|N\|_F^2$  is minimal when



$\|N\|_F = 0$ , i.e., when  $N = 0$ . Therefore,  
 $\hat{\tau} = Y(X^T X)^{-1} X^T$  is the minimal norm solution.